

Dhaka International University



Faculty of Science and Engineering

Department of CSE

Course Name: Data Structure Lab

Course Code: 0613-104

Semester: 2nd **Batch:** 98 (1st Shift)

Roll: 35

Registration Number: CS-D-98-23-127358

Lab Assignment: 01

Assignment on: Array and Searching

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Submitted To:

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Problem: Write a program to insert a value into a sorted list. Location is not given by the user. Use linear search to find the location.

Program Name: Insert in a sorted array using function

Code:

```
#include <stdio.h>

void arrayInput(int a[], int n)
{
    printf("\nEnter array element - \n");
    for(int i=0; i < n; i++)
    {
        printf("index[%d] = ", i);
        scanf("%d", &a[i]);
    }
    printf("\n");
}

void arrayDisplay(int a[], int n)
{
    printf("\nInserted item list - ");
    for(int i=0; i < n; i++)
    {
        printf("\nindex[%d] = %d", i, a[i]);
    }
}

int insert(int a[], int n, int item)
{
    int loc = -1;
    for(int i=0; i < n; i++)
    {
        if(item < a[i])
        {
            loc = i;
            break;
        }
    }
    if(loc < n)
    {
        for (int i = n-1; i >= loc; i--)
        {
            a[i+1] = a[i];
        }
    }
}
```

```

    a[loc] = item;
    n++;

    arrayDisplay(a,n);
    printf("\n\nNew item inserted into index[%d].\n\n", loc);
}

int main()
{
    int n, item;
    printf("\n");
    printf("Enter the number of element : ");
    scanf("%d", &n);

    int a[n];
    arrayInput(a,n);

    printf("\nEnter new item : ");
    scanf("%d", &item);
    insert(a,n,item);

    return 0;
}

```

Output:

```

C/C++ Compile Run - output
> pwsh 5ms 93% 9,22:32
>> cd 'e:\DIU\CSE\2nd Semester\02. Data Structure Lab\Assignment\output'
> pwsh 8ms 93% 9,22:33
>> & .\01. insert in a sorted array using function.exe

Enter the number of element : 5

Enter array element -
index[0] = 10
index[1] = 20
index[2] = 40
index[3] = 50
index[4] = 60

Enter new item : 30

Inserted item list -
index[0] = 10
index[1] = 20
index[2] = 30
index[3] = 40
index[4] = 50
index[5] = 60

New item inserted into index[2].

> pwsh 26s 117ms 93% 9,22:33
>>

```

Program Name: Insert in a sorted array without using function

Code:

```
#include <stdio.h>
int main()
{
    int i, loc, n, e, a[50], item;

    printf("\nEnter No. Of Elements: ");
    scanf("%d", &n);
    printf("\n");

    for(int i=0; i<n; i++)
    {
        printf("index[%d] = ", i);
        scanf("%d", &a[i]);
    }
    for(;;)
    {
        loc = -1;
        printf("\nEnter Inserting Item: ");
        scanf("%d", &item);

        for(int i=0; i<n; i++)
        {
            if(item < a[i])
            {
                loc = i;
                break;
            }
        }
        if(loc<n)
        {
            for(i=n-1; i>=loc; i--)
            {
                a[i+1]=a[i];
            }
            a[loc]=item;
            n=n+1;
            printf("\nList after Insert:\n");

            for(i=0; i<n; i++)
            {
                printf("\nindex[%d] = %d", i, a[i]);
            }
            printf("\nPress any value for exit and 1 for continue: ");
        }
    }
}
```

```

        scanf("%d", &e);
        if(e!=1)
            return 0;
    }
    return 0;
}

```

Output:

```

C/C++ Compile Run - output
> pwsh Assignment 93ms 89% 9,22:40
>> cd 'e:\DIU\CSE\2nd Semester\02. Data Structure Lab\Assignment\output'
> pwsh output 37ms 89% 9,22:40
>> & .\'02. insert in a sorted array.exe'

Enter No. Of Elements: 5

index[0] = 10
index[1] = 20
index[2] = 30
index[3] = 50
index[4] = 60

Enter Inserting Item: 40

List after Insert:

index[0] = 10
index[1] = 20
index[2] = 30
index[3] = 40
index[4] = 50
index[5] = 60
Press any value for exit and 1 for continue: 0
> pwsh output 43s 416ms 89% 9,22:41
>> 

```

Attachment:

1. C code file
2. Output Image